

Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Numerical

An electromagnetic wave of frequency 5GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is $\text{---} \times 10^7 \text{ m/s}$

Q2: 24 Feb (Shift 2) - Single Correct

Match List - I with List - II. List - I List - II

List - I**List - II**

- | | |
|-----------------------------------|---------------------------------------|
| (a) Source of microwave frequency | (i) Radioactive decay of nucleus |
| (b) Source of infrared frequency | (ii) Magnetron |
| (c) Source of Gamma Rays | (iii) Inner shell electrons |
| (d) Source of X-rays | (iv) Vibration of atoms and molecules |
| | (v) LASER |
| | (vi) RC circuit |

Choose the correct answer from the options given below :

- (1) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
- (2) (a) – (vi), (b) – (iv), (c) – (i), (d) – (v)
- (3) (a) – (ii), (b) – (iv), (c) – (vi), (d) – (iii)
- (4) (a) – (vi), (b) – (v), (c) – (i), (d) – (iv)

Q3: 24 Feb (Shift 2) - Numerical

An electromagnetic wave of frequency 3GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be $\text{---} \times 10^{-2} \text{ cm}$

Q4: 25 Feb (Shift 2) - Numerical

The peak electric field produced by the radiation coming from the 8 W bulb at a distance of 10 m is $\frac{x}{10} \sqrt{\frac{\mu_0 c}{\pi}} \frac{\text{V}}{\text{m}}$. The efficiency of the bulb is 10% and it is a point source. The value of x is

Q5: 26 Feb (Shift 1) - Numerical

A radiation is emitted by 1000 W bulb and it generates an electric field and magnetic field at P_t placed at a distance of 2 m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1} \text{ V/m}$. Value of x is

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Answer Key

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Q1 (15)

Q2 (1)

Q3 (667)

Q4 (2)

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Q5 (137)

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Hints and Solutions

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Q1 (15)

Given : $f = 5\text{GHz}$

$$\epsilon_r = 2$$

$$\mu_r = 2$$

$$\text{velocity of wave} \Rightarrow v = \frac{c}{n} \dots (1)$$

where, $n = \sqrt{\mu_r \epsilon_r}$ and $c = \text{speed of light} = 3 \times 10^8 \text{ m/s}$

$$n = \sqrt{2 \times 2} = 2$$

put the value of n in we get

$$\Rightarrow v = \frac{3 \times 10^8}{2} = 15 \times 10^7 \text{ m/s}$$

$$\Rightarrow X \times 10^7 = 15 \times 10^7$$

$$X = 15$$

Q2 (1)

(a) Source of microwave frequency - (ii) Magnetron

(b) Source of infra red frequency - (iv) Vibration of atom and molecules

(c) Source of gamma ray - (i) Radio active decay of nucleus

(d) Source of X-ray - (iii) inner shell electron

Q3 (667)

$$f = 3\text{GHz}, \epsilon_r = 2.25$$

$$V = \lambda f \Rightarrow \lambda = \frac{V}{f}$$

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$V = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} \Rightarrow \lambda = \frac{C}{f \cdot \sqrt{\mu_0 \epsilon_0} \cdot \sqrt{\mu_r \epsilon_r} \cdot f}$$

$$\Rightarrow \lambda = \frac{3 \times 10^8}{f \cdot \sqrt{\mu_r} \cdot \sqrt{\epsilon_r}} \Rightarrow \lambda = \frac{1}{3 \times 10^9 \times \sqrt{1} \times \sqrt{2.25}}$$

$$\Rightarrow \lambda = 667 \times 10^{-2} \text{ cm}$$

Q4 (2)

Hints and Solutions

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$$I = \frac{1}{2} c \epsilon_0 E_0^2$$

$$\frac{8}{4\pi \times 10^2} = \frac{1}{2} \times c \times \frac{1}{\mu_0 c^2} \times E_0^2$$

$$E_0 = \frac{2}{10} \sqrt{\frac{\mu_0 c}{\pi}}$$

$$\Rightarrow x = 2$$

Q5 (137)

Intensity of electro magnetic wave is,

$$I = \frac{1}{2} C \epsilon_0 E_0^2 = \frac{P}{4\pi r^2}$$

$$\frac{1}{2} 4\pi \epsilon_0 \times C \times E_0^2 = \frac{P}{r^2}$$

$$\frac{1}{2} \times \frac{3 \times 10^5 \times E_0^2}{9 \times 10^9} = \frac{1000 \times 1.25}{(2)^2} \times \frac{1}{100}$$

$$E_0^2 = \frac{60 \times 1000 \times 1.25}{4 \times 100} = \frac{125 \times 3}{2}$$

$$E_0^2 = \frac{375}{2} = 187.5$$

$$E_0 = 13.69$$

$$E_0 \approx 137 \times 10^{-1} \text{ V/m}$$